Team Name : Cosmovexa

Team Leader Name : Arpit Tiwari (Ajay kumar garg engineering college ,Ghaziabad)

Problem Statement : Generative AI-Based Cloud Removal and Reconstruction for LISS-IV Satellite Imagery

Team Members

Team Leader:

Name: Arpit Tiwari
College: Ajay kumar garg engineering college ,Ghaziabad

Team Member-1:

Name:Anand Tripathi
College:Ajay kumar garg engineering college ,Ghaziabad

Team Member-2:

Name: Nihal Yadav
College:Ajay kumar garg engineering college ,Ghaziabad

Team Member-3:

Name: Vaishnavi Tyagi
College:Ajay kumar garg engineering college ,Ghaziabad

Opportunity: Why CloudForge Wins

- Differentiation: Most cloud-removal work uses single-image GAN inpainting or simple temporal compositing. CloudForge fuses SAR (cloud-penetrating), temporal optical references, and the cloudy scene via cross-attention, then refines with a diffusion head -- recovering texture, not just plausible color.
- Solving the problem: Hybrid U-Net backbone learns spatial structure; cross-attention injects SAR/temporal context exactly where clouds occlude LISS-IV pixels; the diffusion refinement head sharpens fine-scale detail lost to naive regression-based inpainting.
- USP: Multi-sensor fusion + generative refinement (not masking), operational MLOps pipeline (DVC + MLflow) for reproducibility, and a live Streamlit dashboard for instant before/after, SSIM/PSNR/SAM-based QA -- built for deployment, not just a notebook demo.

Key Features

- **Multi-sensor cloud removal:** Fuses LISS-IV, Sentinel-1 SAR, and temporal optical references to reconstruct cloud-occluded regions.

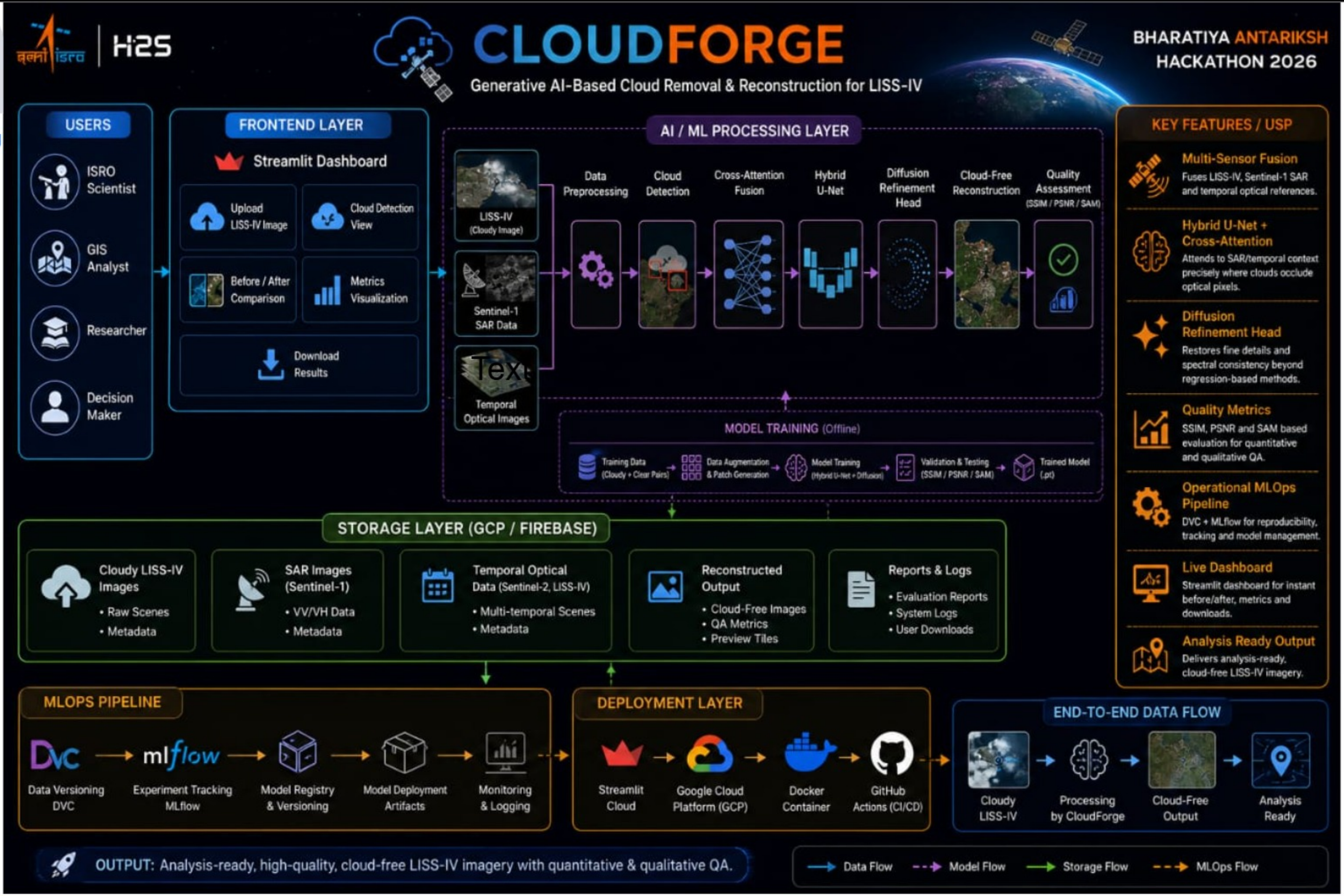
- **Cross-attention fusion:** Hybrid U-Net backbone attends to SAR/temporal context precisely where optical pixels are cloud-contaminated.

- **Diffusion refinement head:** Sharpens fine textures and restores spectral consistency beyond what regression-only models achieve.

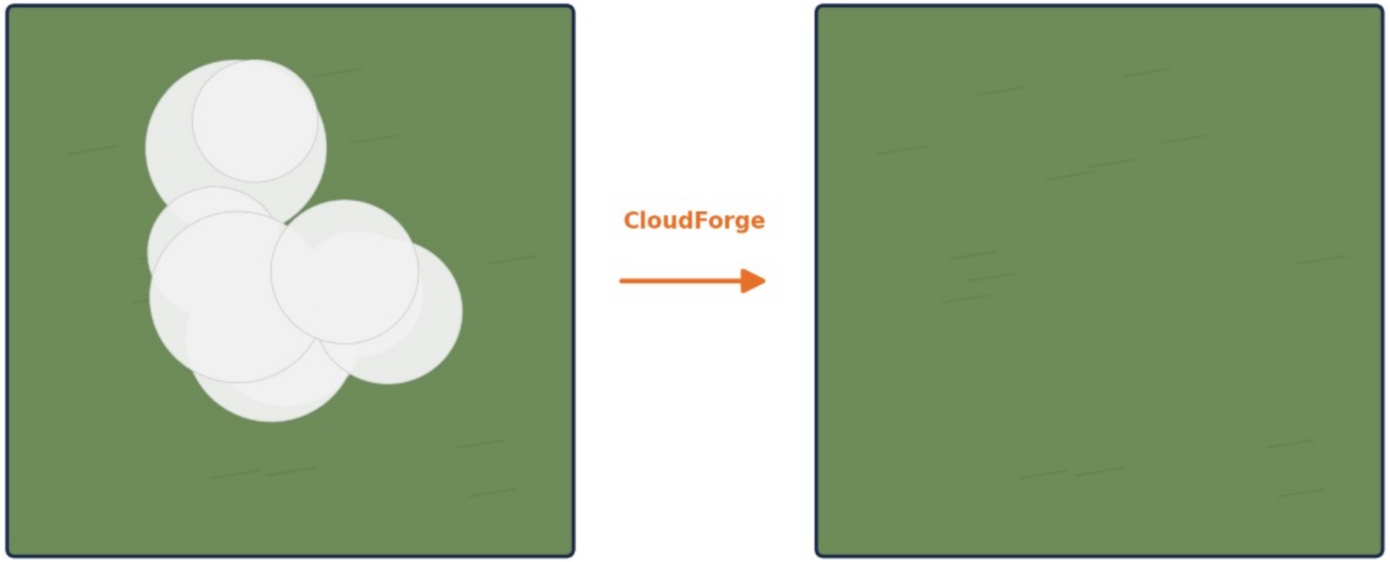
- **Quantitative + qualitative QA:** SSIM, PSNR, and SAM-based evaluation alongside visual before/after comparison.

- **Operational deployment ready:** DVC + MLflow pipeline with a live Streamlit dashboard for monitoring and demo.

Flow Diagram



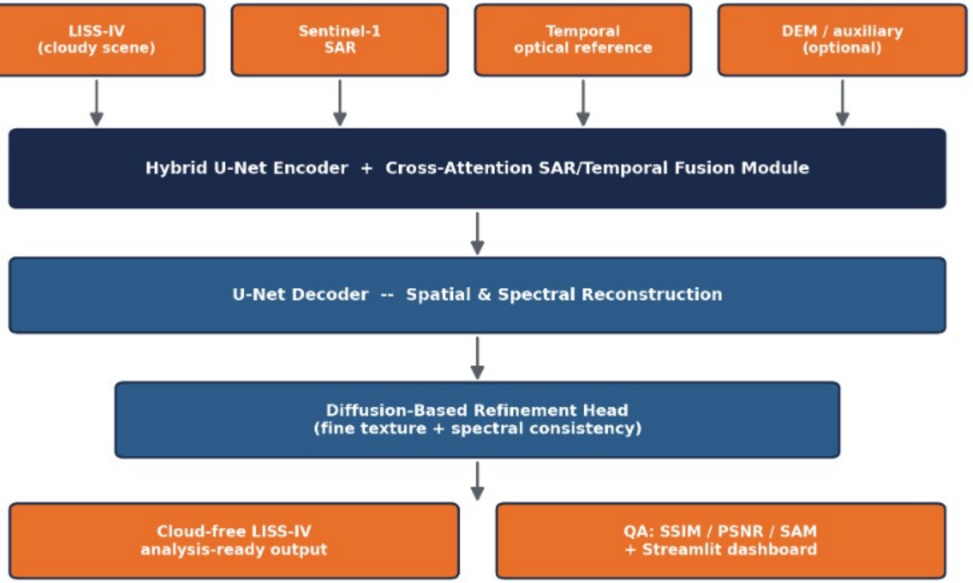
Expected Output: Before / After

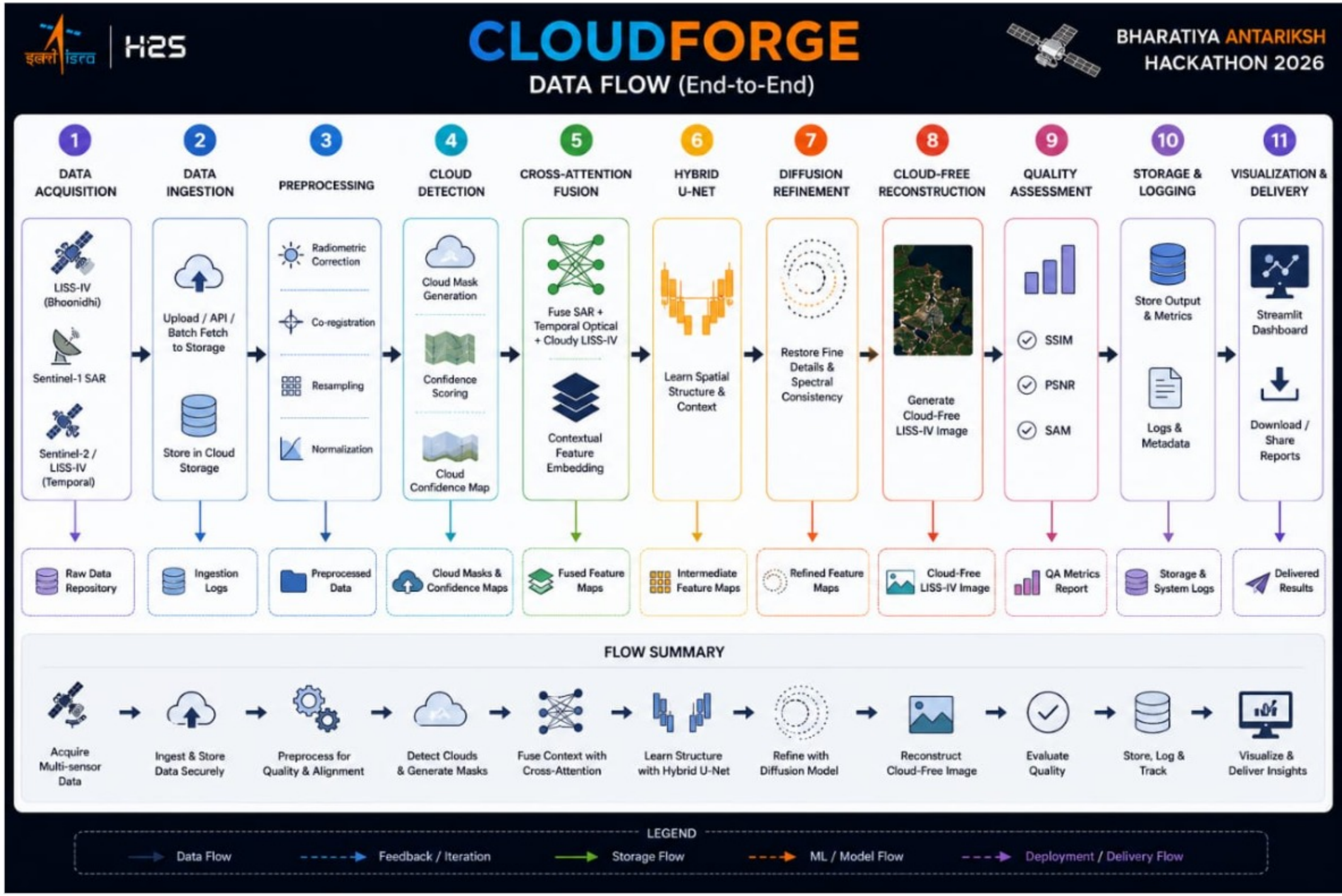


Input: cloudy LISS-IV scene

Output: reconstructed, cloud-free

Architecture: CloudForge





Technology Stack

- **Modeling:** Python, PyTorch, Hybrid U-Net + cross-attention fusion, diffusion-based refinement head
- **Geospatial:** GDAL, Rasterio, QGIS, Google Earth Engine (auxiliary data)
- **Supporting libraries:** OpenCV, NumPy, Scikit-image, Albumentations
- **MLOps & deployment:** DVC, MLflow, Streamlit dashboard for live demo and QA visualization

Estimated Implementation Cost

- **Data:** Free -- LISS-IV via Bhoonidhi, Sentinel-1/2 via Copernicus Open Access Hub, DEM via public sources.
- **Compute:** Single-GPU training feasible (e.g. T4/A100-class); estimated cloud GPU cost for prototyping: under ₹15,000 for the hackathon build phase.
- **Tooling:** Open-source throughout (PyTorch, GDAL, DVC, MLflow, Streamlit) -- zero licensing cost.
- **Scaling to operations:** Marginal cost driven by inference compute only -- diffusion refinement step is the main cost lever and can be made optional for lighter-weight batch runs.

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THANK YOU